Bayesian quantile-based correction and synthesis for climate products

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Code and reproducibility

github.com/AntonioAPDL/Project1

CONTEXT

Archives and ensembles

Recent climate analysis and forecasting rely heavily on climate-center model archives and ensemble prediction systems. **Retrospective products** replay a center's physical–numerical modeling system over past conditions, so simulated flows can be compared with observations and source bias can be learned. **Forecast ensembles** run the operational system from current initial conditions to issue many plausible futures; their spread is useful, but biases, horizons, and update cycles differ by source.

QUESTION

Correct first, then synthesize

Can quantile-level bias be learned from diverse climate-product archives and propagated into newly issued ensemble forecasts on operational update timescales, while preserving an interpretable dynamic decomposition of source corrections, trend, seasonality, and exogenous effects?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge on the $\log(1 + x)$ scale. Before each forecast origin, observed flow is paired with retrospective ECMWF/GloFAS and NOAA/NWS products to estimate source corrections. After the origin, issued ensembles and staged exogenous covariates are used to carry those corrections into source-adjusted quantile forecasts and one posterior predictive distribution.

USGS observations
15-min public gauge record aggregated to daily log-flow; state-updating and held-out verification target.

ECMWF / GloFAS
European Centre for Medium-Range Weather Forecasts; retrospectives; daily 51-member ensemble forecasts, leads 1–28.

NOAA / NWS
U.S. National Weather Service under NOAA; retrospectives; hourly 7-member ensemble forecasts, daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION

Rolling-origin holdout design

Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent extended Dynamic Quantile Linear Model (exDQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. Posterior predictive samples are generated for each fitted quantile lane and synthesized into one posterior predictive distribution.

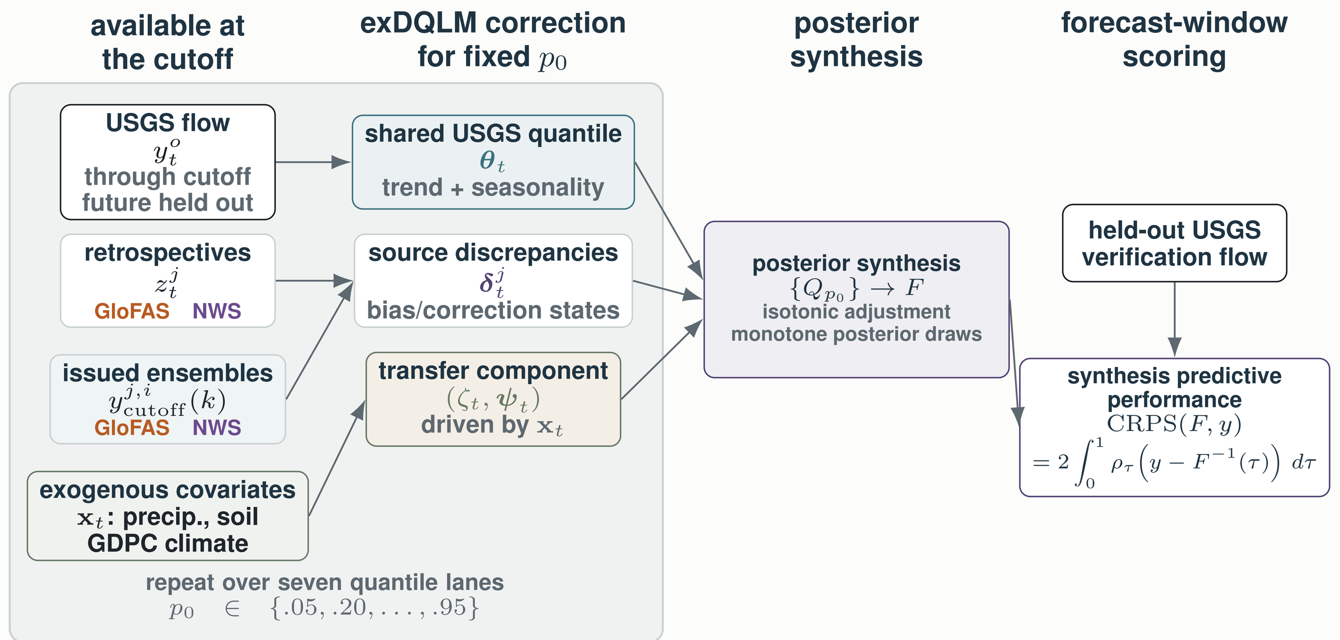
7	$\leq$12,995	~2–4 h
quantile lanes	USGS archive days	fit + synthesis

Parallel quantile lanes
Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

Scalable VB updates
Mean-field variational Bayes supports Kalman filtering and backward smoothing for state-distribution updates. Laplace–Delta approximations handle the non-conjugate exAL scale and skewness parameters.

MODEL

Main Workflow



Source-aware exDQLM workflow. For each target quantile p_0, pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution using isotonic adjustment and monotone rearranged draws, then scored only with held-out USGS observations.

extended Dynamic Quantile Linear Model (exDQLM)

For each target quantile p_0, let T denote the forecast origin/cutoff. The selected exDQLM links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Source discrepancies learn pre-origin corrections; forecast ensembles and staged covariates then propagate corrected quantiles after the origin.

$$\textbf{Observed flow:}\quad y_t^o \sim \mathrm{exAL}_{p_0}(\mathbf{F}_t'\boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o)\,, \qquad 1 \leq t \leq T,$$

$$\textbf{Retrospectives:}\quad z_t^j \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_t'(\boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j) + \zeta_t, \sigma^j, \gamma^j\right), \qquad j \in \mathcal{J},\ 1 \leq t \leq T,$$

$$\textbf{Forecast ensembles:}\quad y_T^{j,i}(k) \sim \mathrm{exAL}_{p_0}\left(\mathbf{F}_{T+k}'(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j) + \zeta_{T+k}, \sigma^j, \gamma^j\right), \ j \in \mathcal{J},\ i \in \mathcal{I}_j,\ k \in \mathcal{K}_j.$$

$$\textbf{Trend/seasonal:}\quad \boldsymbol{\theta}_t \mid \boldsymbol{\theta}_{t-1} \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t^\theta),$$

$$\textbf{Discrepancies:}\quad \boldsymbol{\delta}_t^j \mid \boldsymbol{\delta}_{t-1}^j \sim \mathcal{N}(\mathbf{G}_t\boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{\delta^j}),$$

$$\textbf{Transfer:}\quad \zeta_t \mid \zeta_{t-1}, \boldsymbol{\psi}_{t-1} \sim \mathcal{N}(\lambda\zeta_{t-1} + \mathbf{x}_t'\boldsymbol{\psi}_{t-1}, w_t^\zeta),$$

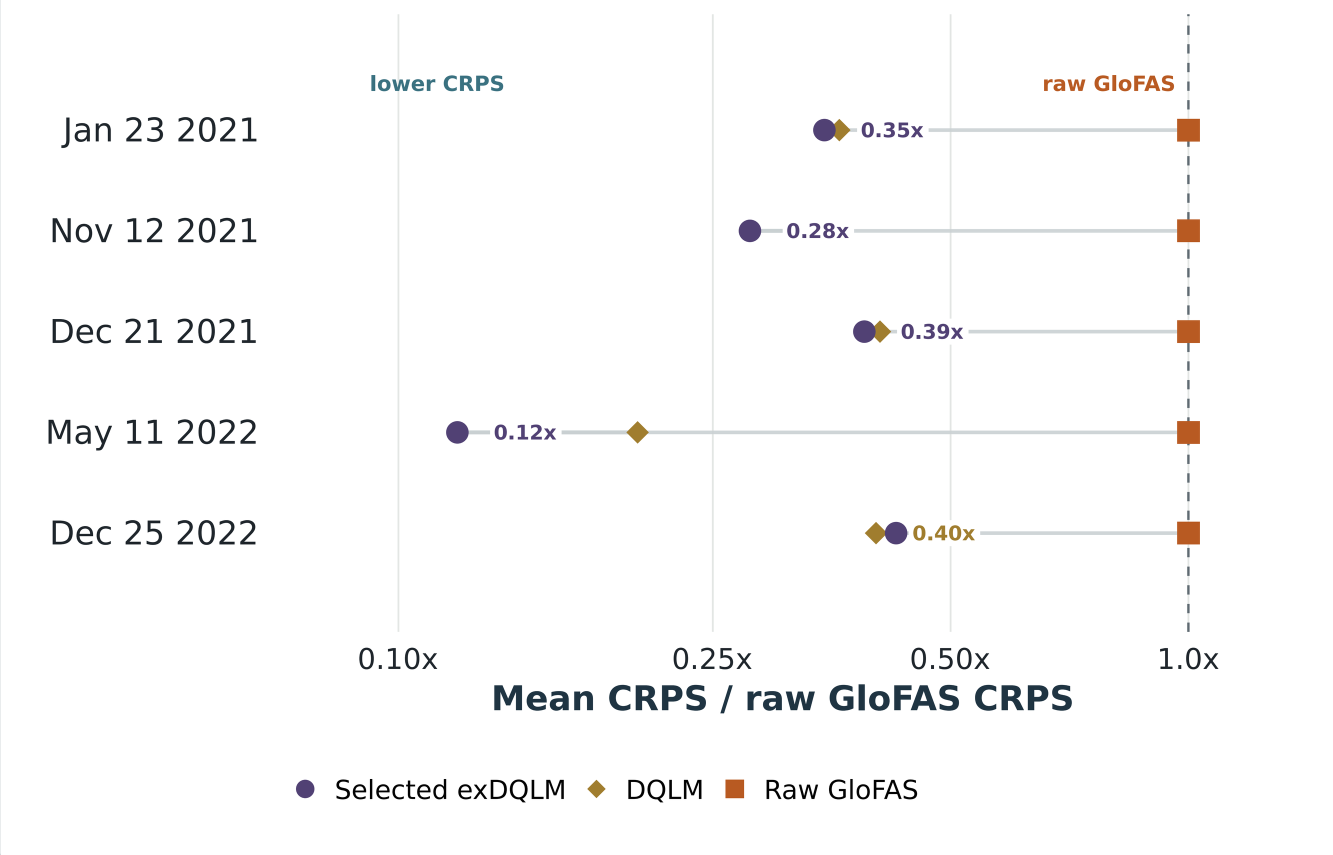
$$\textbf{Covariate effects:}\quad \boldsymbol{\psi}_t \mid \boldsymbol{\psi}_{t-1} \sim \mathcal{N}(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^\psi).$$

Here $\boldsymbol{\theta}_t$ is the shared trend–seasonal river quantile, $\boldsymbol{\delta}_t^j$ is the source-specific correction, and $(\zeta_t, \boldsymbol{\psi}_t)$ is the exogenous-covariate block driven by $\mathbf{x}_t$ (local precipitation, local soil moisture, and GDPC climate summaries). **Priors:** $\boldsymbol{\theta}_0 \sim \mathcal{N}(\mathbf{m}_0, \mathbf{C}_0)$, $\sigma^s \sim IG(a_s, b_s)$, and $\gamma^s \sim t_{(L,U)}(0, \phi^s, \nu^s)$; component discounting controls pre-cutoff state covariances $\mathbf{W}_t$. In the forecast window, covariance is regularized by an Inverse Wishart prior $\mathrm{IW}(\nu_t, \mathbf{S}_t)$, with $\nu_t = p_t + 1 + \epsilon$ and $\mathbf{S}_t = (\nu_t - p_t - 1)c\,\mathbf{W}_T$, centering the prior mean on $c\,\mathbf{W}_T$ after restricting to active forecast states.

RESULTS

1–28-step-ahead forecast CRPS comparison

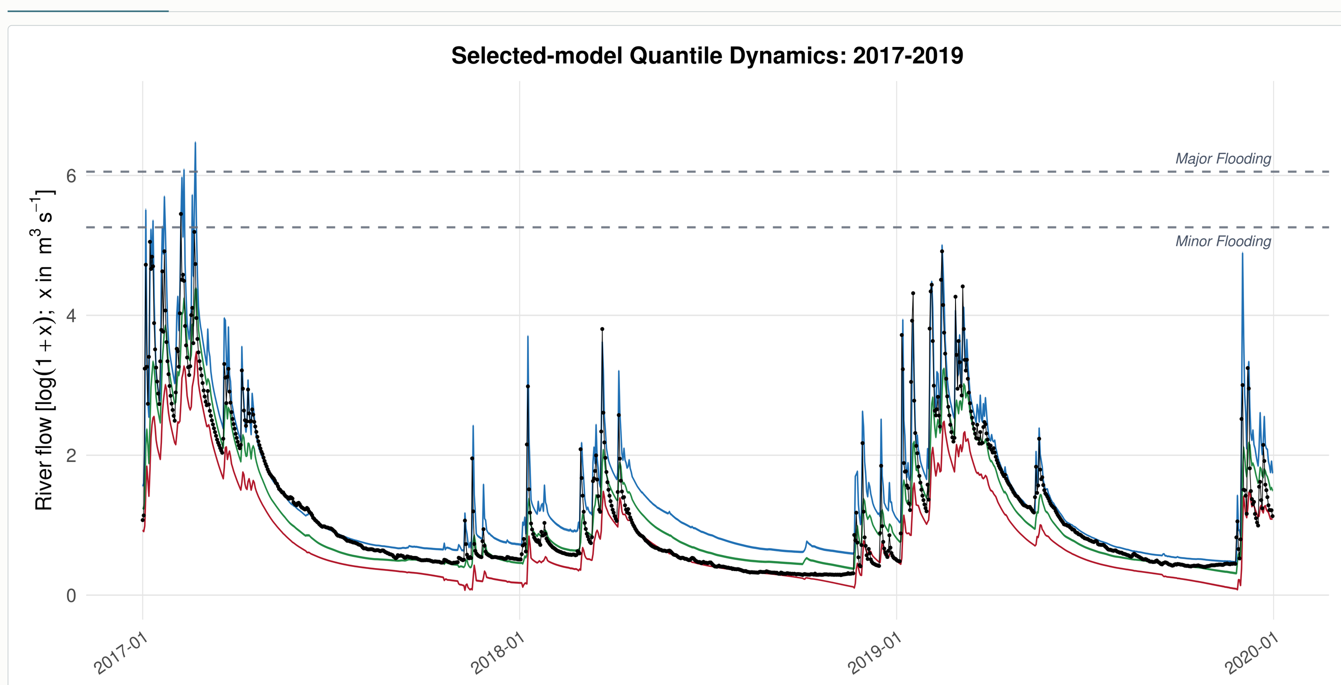
Selected exDQLM is lowest at four of five origins; DQLM leads at 2022-12-25 CRPS ↓



CRPS is the continuous ranked probability score; lower values are better. DQLM is the conjugate AL-likelihood baseline, while selected exDQLM denotes the exAL extension. Raw NWS is evaluated separately over common leads 1–8, not mixed into this 1–28 lead GloFAS-supported benchmark.

FIT

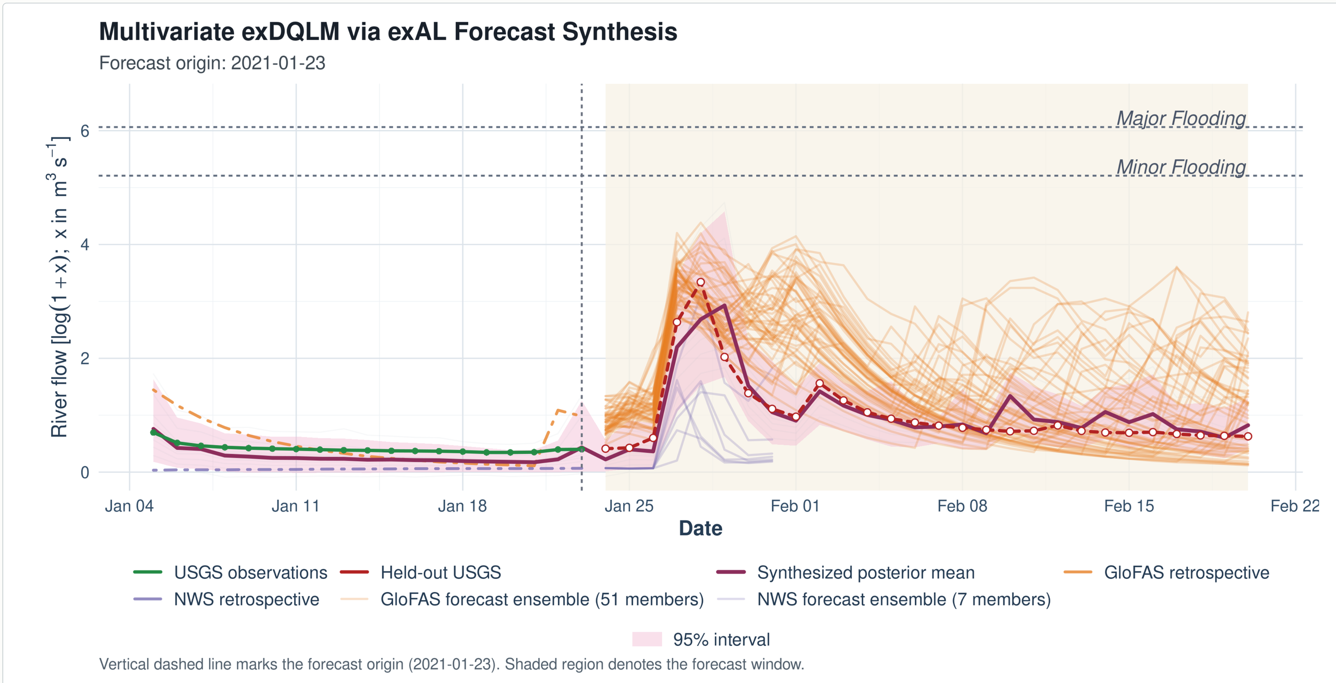
Wet-period quantile fit

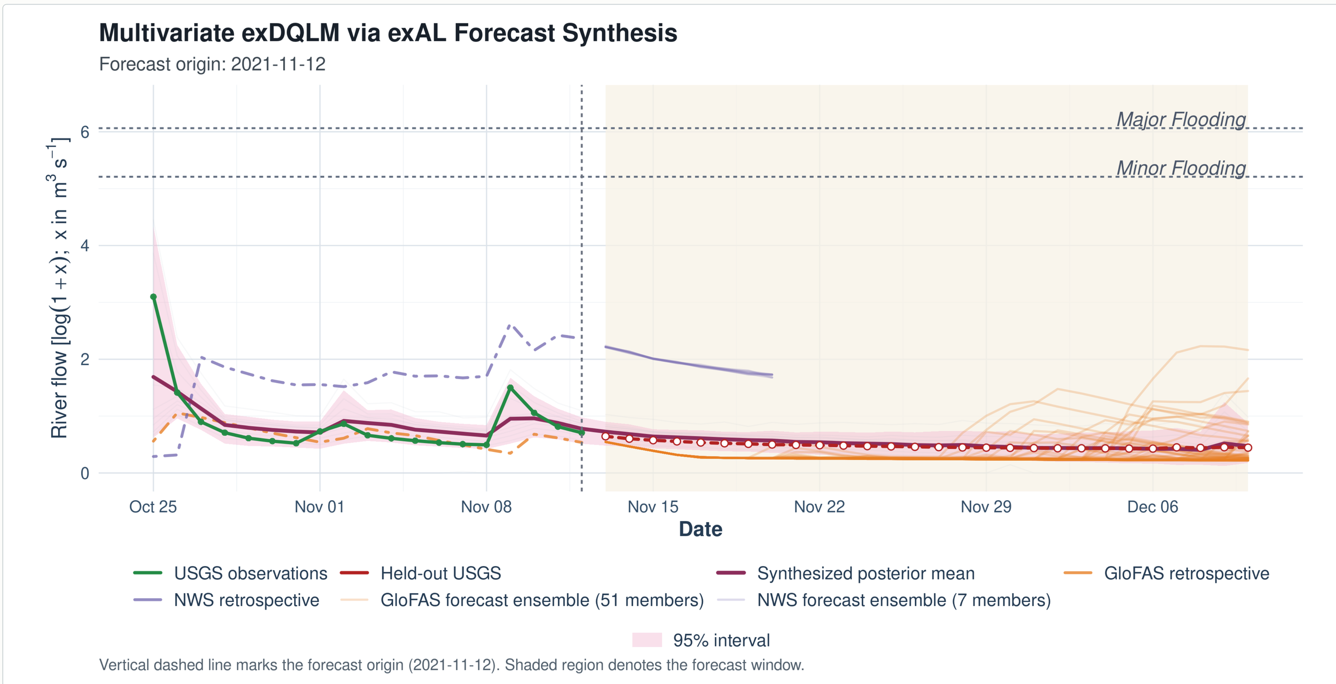


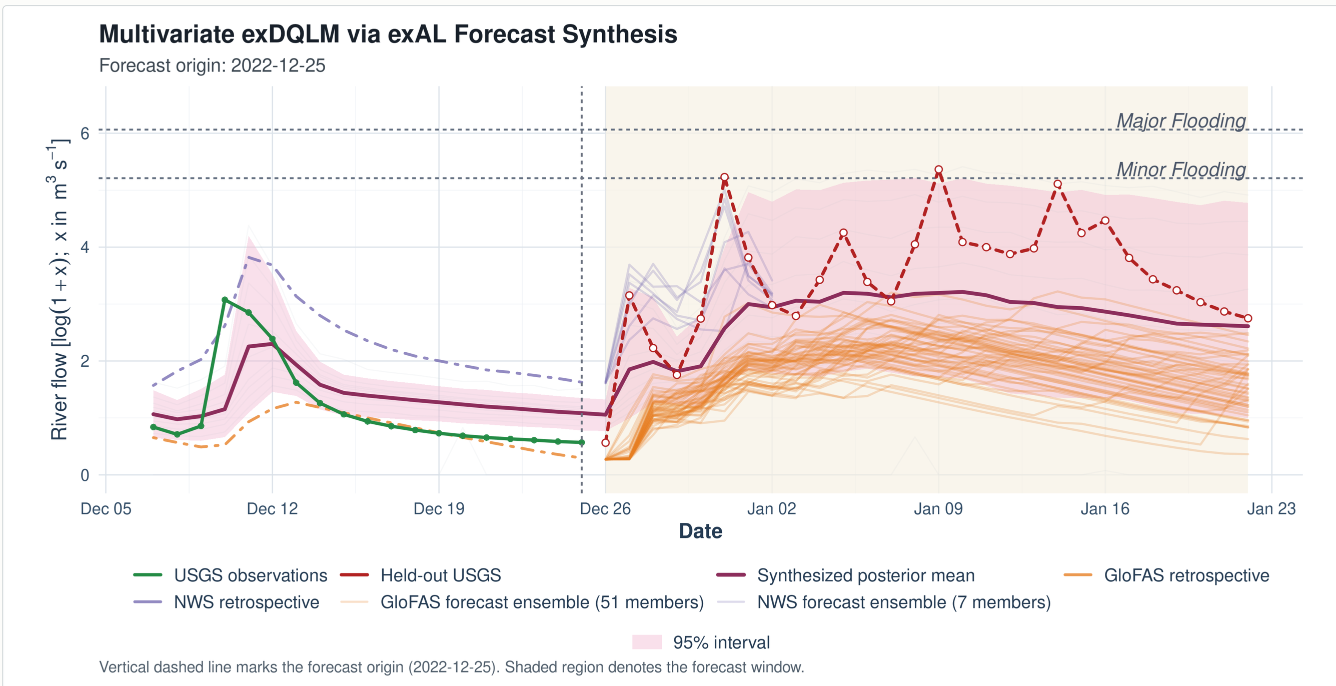
Historical fit diagnostic. Selected exDQLM fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

EXAMPLE

Three forecast-window examples



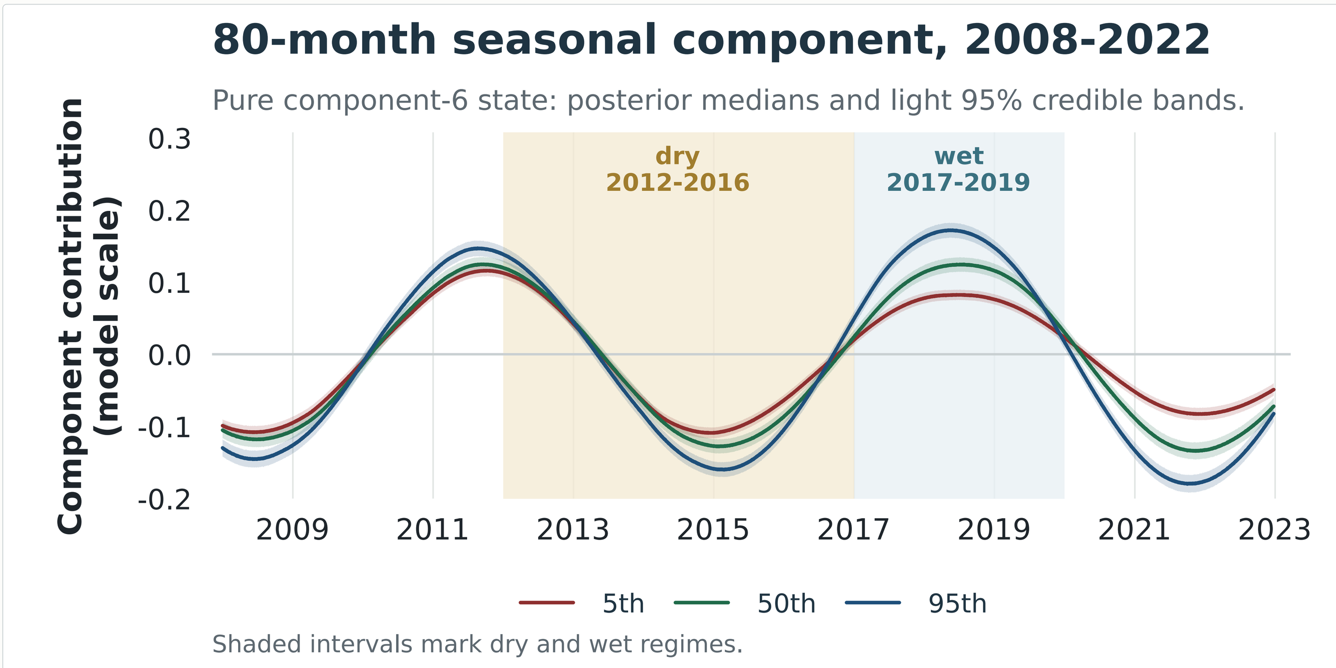




Origin-level illustrations. Three selected exDQLM forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS

Interpretable seasonal component



Retrospective diagnostic. One component of the interpretable state decomposition. From the selected exDQLM at the 2022-12-25 origin, lines show the 80-month seasonal state for the 5th, 50th, and 95th target-quantile lanes over 2008–2022; light ribbons show 95% credible bands. Shaded dry (2012–2016) and wet (2017–2019) regimes mark Santa Cruz-relevant water-supply contrasts, and the recovered long-cycle state tracks this multi-year behavior.

TAKEAWAYS

Main takeaways

- **Source correction.** Quantile-level bias is learned from retrospective archives and propagated into newly issued ensemble forecasts.
- **Operational synthesis.** Seven corrected quantile lanes are fit in parallel and synthesized into one posterior predictive distribution on update timescales.
- **Interpretable dynamics.** The state decomposition separates source corrections from trend, seasonality, and exogenous effects, linking product bias to river-flow dynamics.

Limitations and next steps

- **Quantile coherence.** Crossing was not material here; joint quantile estimation could improve coherence and possibly forecast skill.
- **Nonlinear response.** Linear state dynamics aid interpretation, but may miss nonlinear hydrologic behavior.
- **Deployment tuning.** Discount factors and transfer rate λ require tuning or model selection before operational use.